

Niranjan S

+91 94896 28143 | niranjan011003@gmail.com | github.com/ninja011003 |

<https://www.linkedin.com/in/niranjansuthagar>

EDUCATION

Sri Manakula Vinayagar Engineering College

B.Tech, Artificial Intelligence and Data Science | 8.2 CGPA | Oct 2021 – Jul 2025

Relevant Coursework : Data Structures & Algorithms, Machine Learning, Deep learning, Computer Vision Distributed Systems, Database Systems

WORK EXPERIENCE

Machine Learning Engineer - Contentstack | Aug 2025 – Present

- **Optimized image ingestion pipeline** by implementing scaled JPEG decoding, eliminating redundant Base64 conversions, and switching to compressed JPEG output, **reducing peak memory usage by 63×**, **processing time by 1.9×**, and **per-call payload size by 25×** while maintaining image quality.
- Built **core components of a multi-tenant AI platform serving 500+ enterprises**, integrating **OpenAI, Gemini, Azure OpenAI, and AWS Bedrock** under a unified abstraction layer.
- Built distributed ingestion pipelines processing **50K–10M+** assets per tenant, implementing asynchronous processing, **Kafka-based event streaming and Elasticsearch** indexing for real-time retrieval and analytics.
- Designed and **implemented a distributed usage metering and billing system** supporting customer, user and product-level credit allocation, enabling accurate accounting, quota enforcement and monetization.
- Developed **real-time analytics pipelines using Celery, Kafka, and Elasticsearch** for scalable event processing and monitoring. Built a custom **distributed rate-limiting system using Redis**.

Machine Learning Engineer Intern - Contentstack | Jan 2025 – Aug 2025

- Contributed to **Brand Kit (core AI product)** across microservices (NestJS, FastAPI) using gRPC and REST, shipping multiple production releases.
 - Built and evaluated pipelines for image metadata generation using **AWS Bedrock, GCP Vision, Azure Cognitive Services, and self-hosted models (SAM, CNNs, vision encoders)**.
 - Benchmarked storage systems (**MongoDB vs SurrealDB**) and collaborated with external teams to evaluate scalability and performance.
 - **Resolved 50+ security vulnerabilities** across Python, JavaScript, and Docker; implemented audit logging for critical services. Received multiple recognition awards for the same.
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PROJECTS

Automated Machine Learning Platform

- Built a **no-code AutoML system automating end-to-end ML pipelines** including preprocessing, feature engineering, model training, and evaluation, model selection classifier, information gain based GA.
- Designed a **genetic algorithm with cross-entropy gain-based fitness, information gain-driven crossover, and multiple mutation strategies** for optimal feature selection.
- Implemented a **custom-trained decision tree** for automated model selection, with modular pipelines, cross-validation, and a Streamlit-based UI. **Tech:** Python, Streamlit, Scikit-learn, NumPy, GeneticAlgorithms

Custom Physics Engine

- Implemented a custom 2D **rigid-body physics engine from scratch in Python**, supporting articulated bodies, joint **constraints, collision detection and response, force integration, and rotational dynamics**.
- Built a modular simulation framework with **bodies, joints, motors, actuators, and constraint solvers** to accurately model.
- Optimized the simulation pipeline for real-time execution and seamless integration with ML agents.

Bipedal Locomotion Simulation

- Implemented the NEAT (NeuroEvolution of Augmenting Topologies) algorithm **from scratch to evolve neural network controllers for autonomous bipedal walking**.
 - Integrated the evolutionary framework with the custom physics engine and designed **reward functions** based on forward motion, stability, and energy efficiency.
 - **Trained neural agents over multiple generations** to learn stable walking behaviors without supervised data.
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SKILLS

- Programming: Python, Go, Java, JavaScript / Typescript
- Backend/Systems: FastAPI, Flask, Node.js, Gin, NestJS, gRPC, REST, Microservices, Kafka, Celery, Elasticsearch, Redis, MongoDB, MySQL, SQL, SurrealDB, Docker, Kubernetes, AWS, GCP
- ML/AI: Pytorch, Numpy, TensorFlow, Scikit-learn, CNNs, Keras, Feature Engineering, Langchain